North Bengal International University

Assignment: 01

Assignment Name: Subnetting calculation

Subject : Computer Network

Subject Code : CSE 0612-3232

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Problem Statement: Suppose you are working an ISP company as a network engineer. You have been given an IP address block of 172.17.0.0/20. You must provide for the following subnet arrangement for the company:

(a) 1st subnet, CSE Department LAN, up to 112 hosts
(b) 2nd subnet, EEE Department LAN, up to 210 hosts
(c) 3rd subnet, ETE Department LAN, up to 64 hosts
(d) Point to point link between CSE and ETE,
(e) Point to point link between CSE and EEE,
(f) Point to point to point link between EEE and ETE.

Fill up the following table, configure the exhibit and test the connectivity of the following cases:
(i) Ping from PC0 to PC5 (ii) PC1 to PC3 and (iii) PC2 to PC5

Subnet Name	Needed Size	Allocated Size	Network Address	Mask	Assignable Range	Broadcast Address
CSE						
EEE						
ETE						
CSE-ETE						
CSE-EEE						
EEE-ETE						

Solve: 172.17.0.0/20

20 = 11111111 . 11111111 . 11111100 . 00000000

Step-1: Sort (descending order) the number of subnets:

a. 1st subnet, EEE Department lan = 210+2 = 212
b. 2nd subnet, CSE department lan = 112+2 = 114
c. 3rd subnet, ETE Department lan = 64+2 = 66
d. Point to point link between CSE to ETE = 2+2 = 4
e. Point to point link between CSE to EEE = 2+2= 4
f. Point to point link between EEE to ETE = 2+2= 4

Step-2: 172.17.0.0/20

$2^0=1$ $2^1=2$ $2^2=4$ $2^3=8$ $2^4=16$ $2^5=32$ $2^6=64$ $2^7=128$ $2^8=225$

Host <= 2^n N=8

Step-3:

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1		
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Network of EEE 212	172.17.0.0/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1st host of EEE	172.17.0.1/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	last host of EEE	172.17.0.254/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	Broadcast EEE	172.17.0.255/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0														
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	Network of CSE 114	172.17.1.0/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1st host of CSE	172.17.1.1/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	last host of CSE	172.17.1.126
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	1	Broadcast CSE	172.17.1.127/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0														
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	Network for ETE 66	172.17.1.128/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	1st host for ETE	172.17.1.129/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	last host for ETE	172.17.1.254/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	Broadcast ETE	172.17.1.255/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0														
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	CSE to ETE 4	172.17.2.0/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	1st host	172.17.2.1/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	last host	172.17.2.2/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	Broadcast	172.17.2.3/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0														
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	CSE o EEE 4	172.17.2.4/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	1st host	172.17.2.5/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	last host	172.17.2.6/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	Broadcast	172.7.2.7/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0														
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	EEE to ETE 4	172.17.2.8/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	1	1st host	172.17.2.9/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	last host	172.17.2.10/20
1	0	1	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	1	Broadcast	172.17.2.11/20

Subnet Name	Needed Host	Allocate Host	Network Address	Mask	Assigned Range	Broadcast Address
EEE	210	212	172.17.0.0/20	255.255.255.0	172.17.0.1 - 172.17.0.254	172.17.0.255
CSE	112	114	172.17.1.0/20	255.255.255.128	172.17.1.1 - 172.17.1.126	172.17.1.127
ETE	64	66	172.17.1.128/20	255.255.255.128	172.17.1.129 - 172.17.1.254	172.17.1.255
CSE to ETE	2	4	172.17.2.0/20	255.255.255.252	172.17.2.1 - 172.17.2.2	172.17.2.3
CSE to EEE	2	4	172.17.2.4/20	255.255.255.252	172.17.2.5 - 172.17.2.6	172.17.2.7
EEE to ETE	2	4	172.17.2.8/20	255.255.255.252	172.17.2.9 - 172.17.2.10	172.17.2.11

online calculation:

The network 172.17.0.0/20 has 4094 hosts.
Your subnets need 392 hosts.

NAME	HOSTS NEEDED	HOSTS AVAILABLE	UNUSED HOSTS	NETWORK ADDRESS	SLASH	MASK	USABLE RANGE	BROADCAST
EEE	210	254	44	172.17.0.0	/24	255.255.255.0	172.17.0.1 - 172.17.0.254	172.17.0.255
CSE	112	126	14	172.17.1.0	/25	255.255.255.128	172.17.1.1 - 172.17.1.126	172.17.1.127
ETE	64	126	62	172.17.1.128	/25	255.255.255.128	172.17.1.129 - 172.17.1.254	172.17.1.255
CSE to ETE	2	2	0	172.17.2.0	/30	255.255.255.252	172.17.2.1 - 172.17.2.2	172.17.2.3
CSE to EEE	2	2	0	172.17.2.4	/30	255.255.255.252	172.17.2.5 - 172.17.2.6	172.17.2.7
EEE to ETE	2	2	0	172.17.2.8	/30	255.255.255.252	172.17.2.9 - 172.17.2.10	172.17.2.11